U.G.C. NET Exa
COMPUTE
(Solved with Expl

1. **The Boolean function [~(~p∧q)∧~(~p∧~q)] ∨(p∧r) is equal to Boolean function :**
   (a) q (b) p ∧ r
   (c) p ∨ r (d) p

Ans : (d) Given boolean function is equivalent

$\Rightarrow [\overline{(\overline{p}\cdot q)}\cdot\overline{(\overline{p}\cdot\overline{q})}]+p.r$

$\Rightarrow [(p\cdot\overline{q})\cdot(p+q)]+p.r$

$\Rightarrow [(p+\overline{q}q)]+p.r$

$\Rightarrow p(1+r)$

$= p$

2. **Let us assume that you construct ordered tree to represent the compound proposition (~(p ∧ q)) ↔ (~p ∧ ~ q).**
   Then, the prefix expression and post-fix expression determined using this ordered tree are given as .... and .... respectively.
   (a) ↔~∧pq∨~~pq, pq∧~p~q~ ∨↔
   (b) ↔~∧pq∧~p~q, pq∧~p~q~∧↔
   (c) ↔∧pq∨~~ pq, pq∧~p~~q∧↔
   (d) ↔~∧pq∧~p~q, pq∧~p~~q∧↔

Ans : (b)

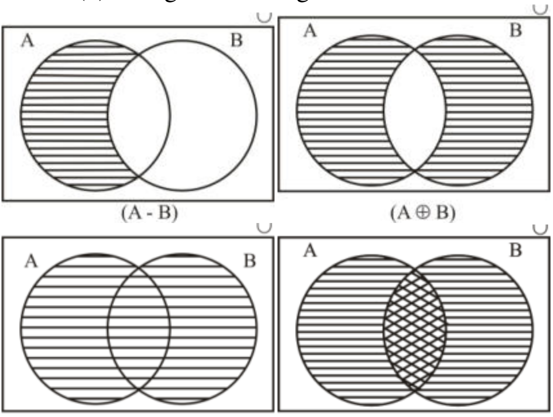
Prefix : ↔ ~ ∧PQV∧~P~Q

Postfix : PQ∧~P~Q~~∨↔

option (b) is answer

3. **Let A and B be sets in a finite universal set U. Given the following :**
   |A–B|, |A ⊕B|, |A|+|B| and |A ∪B|
   Which of the following is in order of increasing size?
   (a) |A–B| ≤ |A ⊕B| ≤ |A|+|B| ≤ |A ∪B|
   (b) |A⊕B| ≤ |A –B| ≤ |A ∪B| ≤ |A|+|B|
   (c) |A⊕B| ≤ |A|+|B| ≤ |A –B| ≤ |A ∪B|
   (d) |A–B| ≤ |A ⊕B| ≤ |A ∪B| ≤ |A|+|B|

Ans : (d) Using Venn's diagram:



(A - B) (A ⊕ B) (A ∪ B) |A| + |B|

Therefore, $|A - B| \le |A \oplus B| \le |A \cup B| \le |A| + |B|$

4. What is the probability that a randomly selected bit string of length 10 is a palindrome?

(a) $\frac{1}{64}$ (b) $\frac{1}{32}$

(c) $\frac{1}{8}$ (d) $\frac{1}{4}$

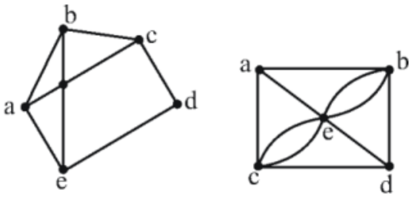
Ans : (b) For palindrome in even length palindrome half length is timed and rest is repeated. So, in 10 bit palindrome, we have 5 position to be filled with 2 choices each i.e. 2^5 choices for first half and 2^5 choices for second half.

0/1	0/1	0/1	0/1	0/1	Fixed	Fixed	Fixed	Fixed	Fixed

Probability = favorable

$$= \frac{2^5}{2^{10}} \Rightarrow \frac{1}{2^5} \Rightarrow \frac{1}{32}$$

5. Given the following graphs :



(G_1) (G_2)

Which of the following is correct?

(a) G_1 contains Euler circuit and G_2 does not contain Euler circuit

(b) G_1 does not contain Euler circuit and G_2 contains Euler circuit

(c) Both G_1 and G_2 do not contain Euler circuit

(d) Both G_1 and G_2 contain Euler circuit

Ans : (c) For a graph to be euler graph every vertex in the graph should have even degree in graph g1 all the vertices have odd degree, so no euler circuit possible therefore g1 is not euler graph.
Graph g1 have vertex a, b, c all these vertex have add degree as 3, 5, 3 so here no euler circuit possible.
So answer is option (c)

6. The octal number 326.4 is equivalent to :

(a) $(214.2)_{10}$ and $(D6.8)_{16}$
(b) $(212.5)_{10}$ and $(D6.8)_{16}$
(c) $(214.5)_{10}$ and $(D6.8)_{16}$
(d) $(214.5)_{10}$ and $(D6.4)_{16}$

Ans : (c) $(326.4)_8 = (3 \times 8^2 + 2 \times 8 + 6 \times 8^0 + 4 \times 8^{-1})_{10} = (214.5)_{10}$
$(326.4)_8 = (011010110.100)_2$
$= (011010110.1000)_2$
$= (OD6.8)_{16}$
$= (D6.8)_{16}$

7. Which of the following is the most efficient to perform arithmetic operations on the numbers?

(a) Sign–magnitude (b) 1's complement
(c) 2's complement (d) 9's complement

Ans : (c) In 1's complement there are two values of zero. A – 0 (11111111) and a + (00000000). On the other hand in 2's complement, there is only one value for 0 (00000000). 2's complement is efficient to perform arithmetic operation on the numbers.

8. The Karnaugh map for a Boolean function is given as

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	0	0	0
$\bar{A}B$	0	0	1	0
AB	1	1	1	1
$A\bar{B}$	0	1	1	1

The simplified Boolean equation for the above Karnaugh Map is

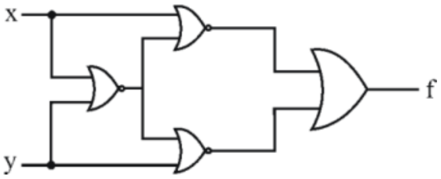
(a) $AB + CD + A\bar{B} + AD$
(b) $AB + AC + AD + BCD$
(c) $AB + AD + BC + ACD$
(d) $AB + AC + BC + BCD$

Ans : (b) Given K map is

AB \ CD	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	1	1	1	1
10	0	1	1	1

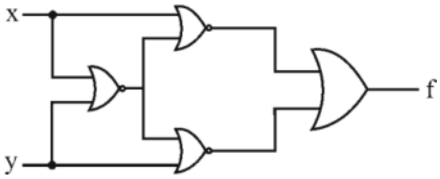
Therefore, boolean function is-
$f = AB + AD + AC + BCD$

9. Which of the following logic operations is performed by the following given combinational circuit?



(a) EXCLUSIVE-OR
(b) EXCLUSIVE-NOR
(c) NAND
(d) NOR

Ans : (a)



f = $\overline{(\overline{(x+y)}+y)}+\overline{(\overline{(x+y)}+x)}$

f = $((\overline{\overline{(x+y)}}\cdot\overline{y}))+((\overline{\overline{(x+y)}}\cdot\overline{x})$

f = $((x+y)\cdot\overline{y})+((x+y)\cdot\overline{x})$

f = $(x\overline{y})+(\overline{x}y)$

f = $x\oplus y = x(\text{EXOR})_y$

10. Match the following :

List - I		List - II	
A.	**Controlled Inverter**	**i.**	**a circuit that can add 3 bits**
B.	**Full adder**	**ii.**	**a circuit that can add two binary numbers**
C.	**Half adder**	**iii.**	**a circuit that transmits a binary word or its 1's complement**
D.	**Binary adder**	**iv.**	**a logic circuit that adds 2 bits**

Codes :

	A	B	C	D
(a)	iii	ii	iv	i
(b)	ii	iv	i	iii
(c)	iii	iv	i	ii
(d)	iii	i	iv	ii

Ans : (d)

- Half adder used to add 2 bits
- Full adder used to add 3 bits
- Controlled inverter is a circuit that transmits a binary word or its d's complement
- Binary adder used to add 2 binary number

Hence answer is (d)

11. Given i = 0, j = 1 k = –1
x = 0.5, y = 0.0
What is the output of given 'C' expression?
x * 3 & & 3 || j I k

(a) –1 (b) 0
(c) 1 (d) 2

Ans : (c) Given C expression with their values-

$= x * y < i + j \,||\, k$

$= 0.5 * 0.0 < 0 + \,||\, 1$

$= 0 < 0 + 1 \,||\, - 1$

$= 0 < 1 \,||\, - 1$

$= 1 \,||\, - 1$

= 1 (returns boolean)

12. The following 'C' Statement :

int * f[] ( ) :

declares :

(a) A function returning a pointer to an array of integers.
(b) Array of functions returning pointers to integers
(c) A function returning an array of pointers to integers
(d) An illegal statement.

Ans : (b) A function returning a pointer to an array of integers

(* f[]) = f is a function returning a pointer then array comes.

Hence the answer is option (b)

13. If a function is friend of a class, which one of the following is wrong?

(a) A function can only be declared a friend by a class itself
(b) Friend functions are not members of a class, they are associated with it
(c) Friend functions are members of a class
(d) It can have access to all members of the class, even private ones

Ans : (c) A friend function of a class is defined outside that class's scope and it has the right to access all private and protected members of the class. Even through the prototypes for friends functions appears in the class definition, friends are not members functions.

14. In C++, polymorphism requires :

(a) Inheritance only
(b) Virtual functions only
(c) References only
(d) Inheritance, Virtual functions and references

Ans : (d) Basically polymorphism means having many forms. In C++, polymorphism requires Inheritance, Virtual functions and references, because polymorphism occurs when there is a hierarchy of classes and they are related by inheritance which include virtual functions and references.

15. A function template in C++ provides .... level of generalization.

(a) 4 (b) 3
(c) 2 (d) 1

Ans : (c) C++ provides two kinds of templates: class templates and function templates. A function template in C++ provides two level of generalization.

16. DBMS provides the facility of accessing data from a database through

(a) DDL (b) DML
(c) DBA (d) Schema

Ans : (b) Accessing the data from the database in DBMS done through DML (Data manipulation language). The DML command can access the data from a database.

17. Relational database schema normalization is NOT for :

(a) reducing the number of joins required to satisfy a query
(b) eliminating uncontrolled redundancy of data stored in the database
(c) eliminating number of anomalies that could otherwise occur with inserts and deletes
(d) ensuring that functional dependencies are enforced

Ans : (a) Relational database schema normalization is not for reducing the number of joins required to satisfy a query.
Relational database schema normalization goal is
(1) eliminating reduclary
(2) function dependencies
So option (a) should be answer

18. Consider the following statement regarding relational database model :

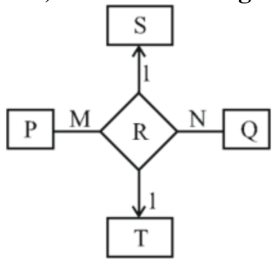
(1) NULL values can be used to opt a tuple out enforcement of a foreign key
(2) Suppose that table T has only one candidate key. If Q is in 3NF, then it is also in BCNF
(3) The difference between the project operator (II) in relational algebra and the SELECT keyword in SQL is that if the resulting table/set has more than one occurrences of the same tuple, then II will return only one of them, while SQL SELECT will return all

One can determine that :

(a) (1) and (2) are true
(b) (1) and (3) are true
(c) (2) and (3) are true
(d) (1), (2) and (3) are true

Ans : (d) (i) In Relational Database Model:
(a) NULL values can be used to opt a tuple out of enforcement of a foreign key correct.
(b) Suppose that table T has only one candidate key. If Q is in 3NF, then it is also in BCNF correct.
(c) The difference between the project operator ($\prod$) in relational algebra and the SELECT keyword in SQL is that if the resulting table/set has more than one occurances of the same tuple, then $\prod$ will return only one of them, while SQL SELECT will return al correct.

19. Consider the following Entity-Relationship (E-R) diagram and three possible relationship sets (I, II and III) for this E-R diagram :



I :

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_1	s_1	t_2

II:

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_1	s_2	t_2

III:

P	Q	S	T
p_1	q_1	s_1	t_1
p_1	q_2	s_1	t_1

If different symbols stand for different values (e.g., t_1 is definitely not equal to t_2), then which of the above could not be the relationship set for the E-R diagram?

(a) I only (b) I and II only

(c) II only (d) I, II and III

Ans : (a) II & II satisfy the 1 : 1 cardinality of S & T but I does not satisfy it
So answer is option (a)

20. Consider a database table R with attributes A and B. Which of the following SQL queries is illegal?

(a) SELECT A FROM R;

(b) SELECT A, COUNT (*) FROM R;

(c) SELECT A, COUNT(*) FROM GROUP BY A;

(d) SELECT A, B, COUNT (*) FROM R GROUP BY A, B;

Ans : (b) SELECT A, COUNT (*) FROM R GROUP By A, B, legal query.
When aggregation function (sum, avg, count, max and min) are used in select clause then we need to use group by clause for grouping the attributes.

21. Consider an implementation of unsorted single linked list. Suppose it has its representation with a head and a tail pointer (i.e. pointers to the first and last nodes of the linked list). Given the representation, which of the following operation can not be implemented in O(1) time?

(a) Insertion at the front of the linked list

(b) Insertion at the end of the linked list

(c) Deletion of the front node of the linked list

(d) Deletion of the last node of the linked list

Ans : (d) Here although we have tail pointer but if will be of no use because these is no previous pointer. We need to here seemed last pointer so that we make it the last pointer & can free the last pointer i.e. deleting the last node. So we need to traverse the whole linked list making it O(n).

22. Consider an undirected graph G where self-loops are not allowed. The vertex set of G is {i, j) $| \; | \leq i \leq 12, 1 \leq j \leq 12$}. There is an edge between (a, b) and (c, d) if $|a - c| \leq 1$ or $|b - d| \leq 1$. The number of edges in this graph is :

(a) 726 (b) 796

(c) 506 (d) 616

Ans : (c) Total number of vertices = 12 * 12 = 144
The graph formed by the decription contains 4 (corner) vertices of degree 3 and 4 o(external) vertices of degree 5 and 100 (remaining/internal) vertices of degree 8.
According to (Handshake theoram)

$2|E| = \text{Sum of degrees}$

$2|E| = 4 \times 3 + 40 \times 5 + 100 \times 8$

$2|E| = 1012$

$|E| = \dfrac{1012}{2} = 506 \text{ edges}$

23. The runtime for traversing all the nodes of a binary search tree with n nodes and printing them in an order is :

(a) O (lg n) (b) O(n lg n)
(c) O(n) (d) (On^2)

Ans : (c) Printing the elements in an order means the element should be printed in sorted order & we know inorder traversal of BST takes gives the sorted order & takes O(n) time. The recurrence involved is :
T(n) = 2T (n/2) + c for balanced BST which on solving gives O(n) and also for skewed tree T(n) = T(n – 1) + c which also on solving gives O(n) time.

24. Consider the following statements :

S_1 : A queue can be implemented using two stacks.

S_2 : A stack can be implemented using two queues.

Which of the following is correct?

(a) S_1 is correct and S_2 is not correct
(b) S_1 is not correct and S_2 is correct
(c) Both S_1 and S_2 are correct
(d) Both S_1 and S_2 are not correct

Ans : (c) Explanation of statement 1: Let us how 2 stack S1 & S2 for our purpose. For enqueue operation, we have 2 cases.
(a) S2 is not empty : In that case, we simply pop from this stack
(b) S2 is empty : in that case first we pop all elements from S1 and push into S2. After pushing all the elements so S1 is correct and as same, simply do for S2. So S2 is also correct.
Hence the answer is option (c).

25. Given the following prefix expression :

*** + 3 + ↑ 3 + 3 3 3**

What is the value of the prefix expression?

(a) 2178 (b) 2199
(c) 2205 (d) 2232

Ans : (c) Prefix expression = *+3+3↑3 + 333
⇒*+3+3∧3(3+3)3
⇒*+3+3∧363
⇒*+3+3(3∧6)3
⇒*+3+37293
⇒*+3(3+729)3
⇒*+3732 3
⇒*(3+732)3
⇒*7333
⇒(735)*3
⇒2205 (ans 3)

26. Which of the following statements is not true with respect to microwaves?

(a) Electromagnetic waves frequencies from 300 GHz to 400 THz
(b) Propagation is line-of-sight
(c) Very high-frequency waves cannot penetrate walls
(d) Use of certain portions of the band requires permission from authorities

Ans : (a) The frequencies are between 300 MHz * 400 GHz and the rest are true

27. In a fast Ethernet cabling, 100 base-TX uses .... cable and maximum segment size is ....

(a) twisted pair, 100 meters
(b) twisted pair, 200 meters
(c) fibre optics, 1000 meters
(d) fibre optics, 2000 meters

Ans : (a) **Fast Ethernet:-** Cabling, 100 Base- Tx uses twisted pair cable and maximum segment size is 100 metres.

28. A network with bandwidth of 10 Mbps can pass an average of 12,000 frames per minute with each frame carrying an average of 10,000 bits. What is the throughput of this network?

(a) 1 Mbps (b) 2 Mbps
(c) 10 Mbps (d) 12 Mbps

Ans : (b) 12000 * 10000/60 = 2 Mbps
So answer is option (b)

29. Match the following :

List - I		**List – II**	
A.	**Session layer**	**i.**	**Virtual terminal software**
B.	**Application layer**	**ii.**	**Semantics of the information transmitted**
C.	**Presentation layer**	**iii.**	**Flow control**
D.	**Transport layer**	**iv.**	**Manage dialogue control**

Codes :

	A	B	C	D
(a)	iv	i	ii	iii
(b)	i	iv	ii	iii
(c)	iv	i	iii	ii
(d)	iv	ii	i	iii

Ans : (a)
(a) Application layer is the topmost layer in TCP/IP protocol stack. Hence it is related to software related things.
(b) Presentation layer present the data in proper format for transmission so it is connected with semantics of data.
(c) Session layer maintains multiple connection and synchrony sation
(d) Lastly transport layer deals with flow control and congession control mainly.
Hence answer is option (a)

30. Which of the following protocols is used by email server to maintain a central repository that can be accessed from any machine?

(a) POP3 (b) IMAP

(c) SMTP (d) DMSP

Ans : (b) IMAP : The basic idea behind IMAP is for all email server to maintain a central repository that can be accessed from any machine

31. The number of strings of length 4 that generated by the regular expression (0^+ 1^+ | 2^+ 3^+)*, where | is an alternation character and {+, *} are quantification characters is :

(a) 08 (b) 09

(c) 10 (d) 12

Ans : (c) Given Reg Exp = (0+ 1+ | 2+ 3+)*

We can assume this expression as (A+| B+)* we can form the string with A(01) or B(23) or both (01, 23) so below are the lo possible 4 char strings

with 01 ⇒ 01111, 0011, 0001, 0101

with 23 ⇒ 2333, 2233, 2223, 2323

using both 01 & 23 ⇒ 0123, 2301

32. The content of the accumulator after the execution of the following 8085 assembly language program, is :

MVI A, 35H

MOV B, A

STC

CMC

RAR

XRA B

(a) 00H (b) 35H

(c) EFH (d) 2FH

Ans : (d)

MUI A, 35 H || Content of accumulator = 35 H

MOV B, A || Content of register B = 35 H

STC || set carry flag = 1

CMC || Complement carry flag = hot (1) =0

RAR

XRA B

(00110101) XOR (0001 1010)

= 0010 1111

= 2F H

33. In compiler optimization, operator strength reduction uses mathematical identities to replace slow math operations faster operations. Which of the following code replacements is an illustration of operator strength reduction?

(a) Replace P + P by 2 * P or Replace 3 + 4 by 7

(b) Replace P * 32 by P < < 5

(c) Replace P * 0 by 0

(d) Replace (P < < 4) – P by P * 15

Ans : (b) Multiplication (*) operation is slower than addition (+) operation and left shift (<<) only replace P* 32 by P<<5 is correct and P*0 by 0 is not envolved any operator reduction.

34. **Which of the following are the principles tasks of the linker?**

I. Resolve external references among separately compiled program units.

II. Translate assembly language to machine code.

III. Relocate code and data relative to the beginning of the program.

IV. Enforce access-control restrictions on system libraries.

(a) I and II (b) I and III

(c) II and III (d) I and IV

Ans : (b) Statement (i) and (iii) are task of linker, and (ii) is task of assembler (iv) using access control.

35. **Which of the following is FALSE?**

(a) The grammar $S \rightarrow aS| aSbS \in ?|$. where S is the only non-terminal symbol, and $\in$ is the null string, is ambiguous

(b) An unambiguous grammar has same left most right most derivation

(c) An ambiguous grammar can never be LR (k) for any k.

(d) Recursive descent parser is a top-down parser

Ans : (b) Recursive descent is a top-down parser is true An unambiguous grammar has some left most & right most derivation True.

The grammar $S \rightarrow aS| aSbS \in$, where S is the only non terminal symbol, & EE is the null string, is ambiguous try to generate aab

Answer is option (b)

36. **Consider a system with seven processes A through G and six resources R through W. Resource ownership is as follows :**

process A holds R and wants T

process B holds nothing but wants T

process C holds nothing but wants S

process D holds U and wants S & T

process E holds T and wants S

process F holds W and wants S

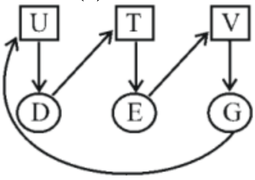
process G holds V and wants U

Is the system deadlocked? If yes, ... processes are deadlocked.

(a) No (b) Yes, A, B, C

(c) Yes, D, E, G (d) Yes, A, B, F

Ans : (c)



System is deadlock due to process D, E, G

So answer is option (c)

37. **Suppose that the virtual address space has eight pages and physical memory with four page frames. If LRU page replacement algorithm is used, ... number of page faults occur with the reference string.**

0 2 1 3 5 4 6 3 7 4 7 3 3 5 5 3 1 1 1 7 2 3 4 1

(a) 11 (b) 12
(c) 10 (d) 9

Ans : (*)

0	2	1	3	5	4	6	3	7	4	7	3	3	5	5	3	1	1	1	7	2	3	4	1
0	0	0	0	5	5	5		7					7			7				7		7	1
	2	2	2	2	4	4		4					4			1				1		4	4
		1	1	1	1	6		6					5			5				2		2	2
			3	3	3	3		3					3			3				3		3	3

Total number of page fault = 13
So, no option is matching

38. Consider a system having 'm' resources of the same type. These resources are shared by three processes P_1, P_2 and P_3 which have peak demands of 2, 5 and 7 resources respectively. For what value of 'm' deadlock will not occur?

(a) 70 (b) 14
(c) 13 (d) 7

Ans : **(b)** Peak demands of P1, P2 & P3 are 2, 5 & 7 allocate 1 less than peak dam and i.e. P1 $\rightarrow$1, P2$\rightarrow$4 P3$\rightarrow$6
The number of resources for which deadlock occurs is 1 + 4 + 6 = 11
Add one to the number of resources we get 12, the value for which deadlock will not occur.

39. Five jobs A, B, C, D and E are waiting in Ready Queue. Their expected runtimes are 9, 6, 3, 5 are x respectively. All jobs entered in Ready queue at time zero. They must run in .... order to minimize average response time if 3 < x < 5.

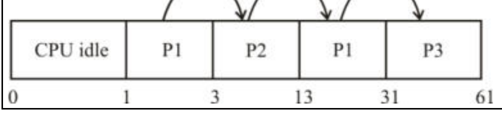
(a) B, A, D, E, C (b) C, E, D, B, A
(c) E, D, C, B, A (d) C, B, A, E, D

Ans : **(b)** To minimize response time these must run in shortest job first order.
Therefore, C(3), E($3 < x < 5$), D(5), B(6), A(9)

40. Consider three CPU intensive processes P1, P2, P3 which require 20, 10 and 30 units of time, arrive at times 1, 3 and 7 respectively. Suppose operating system is implementing shortest Remaining time first (preemptive scheduling) algorithm, then .... context switches are required (suppose context switch at the beginning of ready queue and at the end of Ready queue are not counted).

(a) 3 (b) 2
(c) 4 (d) 5

Ans : **(a)** P1, P2, P3 are three processes
Arrival time of P1 is 1 so we made to run P1 at time 1 now P1 is run for 2 units, process arrived with less B.T, so switch happens to P2-(1) P2 run till completion since there are no shorter Jobs available, then switch happens to P1 ....(2)
Again P1 is run since it has burst time (is) < burst time of P3 (30) completion of P1, switch happens to P3 run till completion..........(3)



41. Which of the following is used to determine the specificity of requirements?

(a) $\frac{n_1}{n_2}$ (b) $\frac{n_2}{n_1}$

(c) $n_1 + n_2$ (d) $n_1 - n_2$

Where n_1 is the number of requirements of which all reviewers have identical interpretations, n_2 is number of requirements in a specification.

Ans : (a) Choice 1 & 2 must be different assuming choice 2 as (na| n1). So the answer is option (a)

42. The major shortcoming of waterfall model is :

(a) the difficulty in accommodating changes after requirement analysis
(b) the difficult in accommodating changes after feasibility analysis
(c) the system testing
(d) the maintenance of system

Ans : (a) The main drawback of the waterfall model is the difficulty of accommodating change after the process is under way.
(1) **Difficult to address change:** Inflexible partitioning of the project into distinct stage makes it difficult to respond to changing customer requirements. Therefore, this model is only appropriate when the requirement are well understood and changes will be fairly limited during the design process
So answer is option (a)

43. The quick design of a software that is visible to end users leads to ....

(a) iterative model (b) prototype model
(c) spiral model (d) waterfall model

Ans : (b) Prototype typically simulates only a few aspects of, and may be completely different from, the final product.
Prototyping has several benefits. The software designer and implementer can get valuable feedback from the user early in the project.

44. For a program of k variables, boundary value analysis yields .... test cases.

(a) $4k - 1$ (b) $4k$
(c) $4k + 1$ (d) $2^k - 1$

Ans : (c) A program of k variables boundary value analysis yields 4*k + 1 test Robustness testing yields 6*k + 1 worst case testing yields 5*k.

45. The extent to which a software performs its intended functions without failures, is termed as :

(a) Robustness (b) Correctness
(c) Reliability (d) Accuracy

Ans : (c) Reliability is the probability of failure free operation of a system over a specified time within a specified environment for a specified purpose.

46. An attacker sits between the sender and receiver and captures the information and

retransmits to the receiver after some time without altering the information. This attack is called is ....
(a) Denial of service attack
(b) Masquerade attack
(c) Simple attack
(d) Complex attack

Ans : (a) Simple and complex attack : Heard first time of this type of attack, classification is actually active and passive attack.
Denial of service attack : Not giving user to access his resource which he is supposed to use
Masquerade attack : Using fake identity to gain authorized access
So answer is option (a)

47. .... is subject oriented, integrated, time variant, nonvolatile collection of data in support of management decisions
(a) Data mining
(b) Web mining
(c) Data warehouse
(d) Database management system

Ans : **(c)** Data warehouse is a subject oriented integrated, time variant and non volatile collection of data in support of management decision making process. A data warehouse is a copy of transaction data specifically structured for query & analysis.

48. In data mining, classification rules are extracted from ....
(a) Data (b) Information
(c) Decision tree (d) Database

Ans : **(c)** A decision tree can be used to visually and explicitly represent decision & division making. In data mining, a decision tree describes data but not decisions; rather the resulting classification tree can be an input for decision making.

49. Discovery of cross sales opportunities is called as ....
(a) Association (b) Visualization
(c) Correlation (d) Segmentation

Ans : **(a)** Association rule learning is a method for discovering intersecting relations between variables in large data base. It is intended to identify strong rules disconnected in database using some measures of interestingness.

50. In Data mining, ... is a method of incremental conceptual clustering.
(a) STRING (b) COBWEB
(c) CORBA (d) OLAD

Ans : (b) COBWEB is an in rental system for hierarchal conceptual clustering. COBWEB incrementally organizes observation into a classification tree. Each nod in classification tree represents a class (concept) and is labeled by a probalistic concept that summarizes the attribute value distribution of objects classified undo the node.